🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.4

Describe the Data Using the Median

Lesson 2-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes `Level 2 Standard`

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the mean, median, and mode of a data set.
Language Objective: I can explain my reasoning using the words mean, median, mode, and outlier.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean, Median, and Mode | **Mean** | **Median** | **Mode** | **Outlier** | **Data distribution** | **Variability**

👆 *Tap any word to see what it means and a picture.*

1. ______________ — Measures that describe a data set's center: average, middle value, and most frequent value.

2. The average found by adding all values and dividing by how many there are is the ______________.

3. The middle value of a data set when the numbers are in order is the ______________.

4. The value that appears most often in a data set is the ______________.

5. A value that is much larger or smaller than the rest of the data is an ______________.

6 The way data values are spread out is the ______________.

7 How spread out the data values are is the ______________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Who is using the mean, and who is using the mode? Which is more useful here?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Mean, Median, and Mode** — Measures that describe a data set's center: average, middle value, and most frequent value.
  Media, mediana y moda — Medidas que describen el centro de un conjunto de datos: promedio, valor central y valor más frecuente.

- [ ] **Mean** — The average. Add all the numbers, then divide by how many there are.
  Media — El promedio. Suma todos los números y divide entre cuántos hay.

- [ ] **Median** — The middle number when you put them in order.
  Mediana — El número del medio cuando los pones en orden.

- [ ] **Mode** — The number that shows up most often.
  Moda — El número que aparece más veces.

- [ ] **Outlier** — A number that is much bigger or smaller than the rest.
  Valor atípico — Un número mucho mayor o menor que los demás.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The teacher used the ____ (about 86.6). The student used the ____ (92). The ____ is more useful here because ____.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Mean, median, and mode each describe the center of the data in a different way.

Evidence: Students distinguish the three measures and justify a choice, noting the mode (38) is the most repeated and the mean/median sit near the center.

Reasoning: This evidence connects the definition of median to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The teacher used the ____ (about 86.6). The student used the ____ (92). The ____ is more useful here because ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean, Median, and Mode
2. **Notes 2:** Mean
3. **Notes 3:** Median
4. **Notes 4:** Mode
5. **Notes 5:** Outlier
6. **Notes 6:** Data distribution
7. **Notes 7:** Variability
8. **Watch for this mistake:** *A common mistake in Mean, Median, and Mode is finding the median without ordering the data first — students pick the middle number straight from the list as given (for 9, 4, 7, 2, 6 they say the median is 7, the middle position in that original order) instead of sorting least to greatest (2, 4, 6, 7, 9) and finding the true middle value, 6. Before finalizing a median, always reorder the data smallest to largest, then count to the middle position — and if there is an even number of values, average the two middle numbers.*
9. **Try It 1:** A. 4 — *The mode is the value that appears most often. 4 shows up twice; every other value appears once. So the mode is 4.*
10. **Try It 2:** A. The median, because the outlier barely moves the middle value — *A high outlier pulls the mean upward, making it larger than most games. The median is the middle of the ordered data, so one unusual game barely changes it — it best describes a typical game.*